Cloud Security Configuration-Lab – AWS Free Tier

Aim

To implement cloud security configuration using AWS Free Tier.

Note: Based on account type and permission levels, certain configurations may not be accessible to all users. The steps should be adapted based on available access.

Tools Required

AWS Free Tier Account

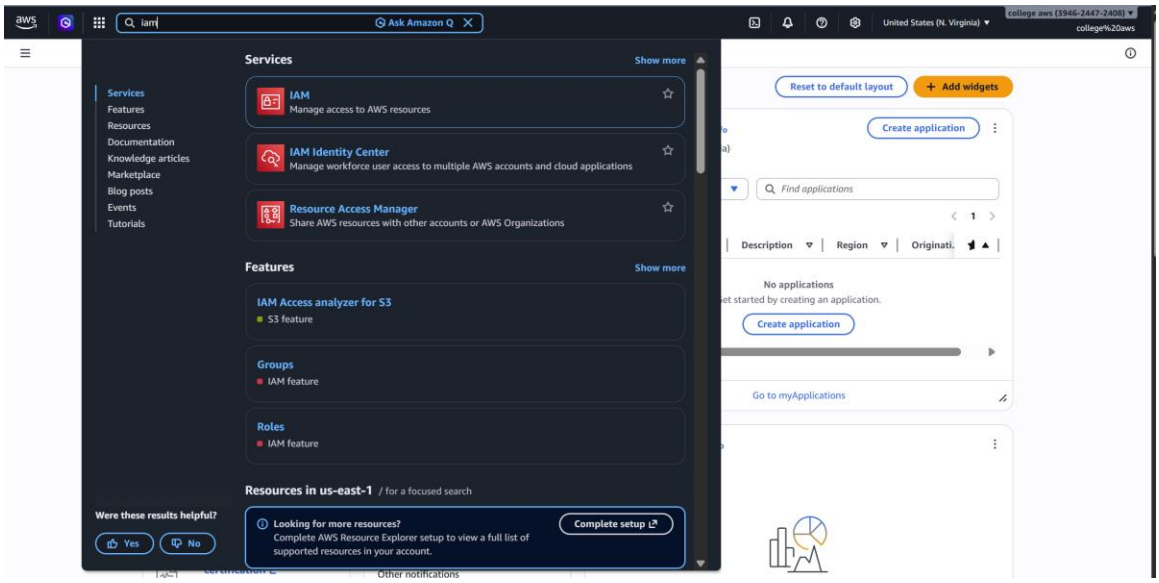
Laptop/Desktop with Internet

Web Browser (Chrome/Edge/Firefox)

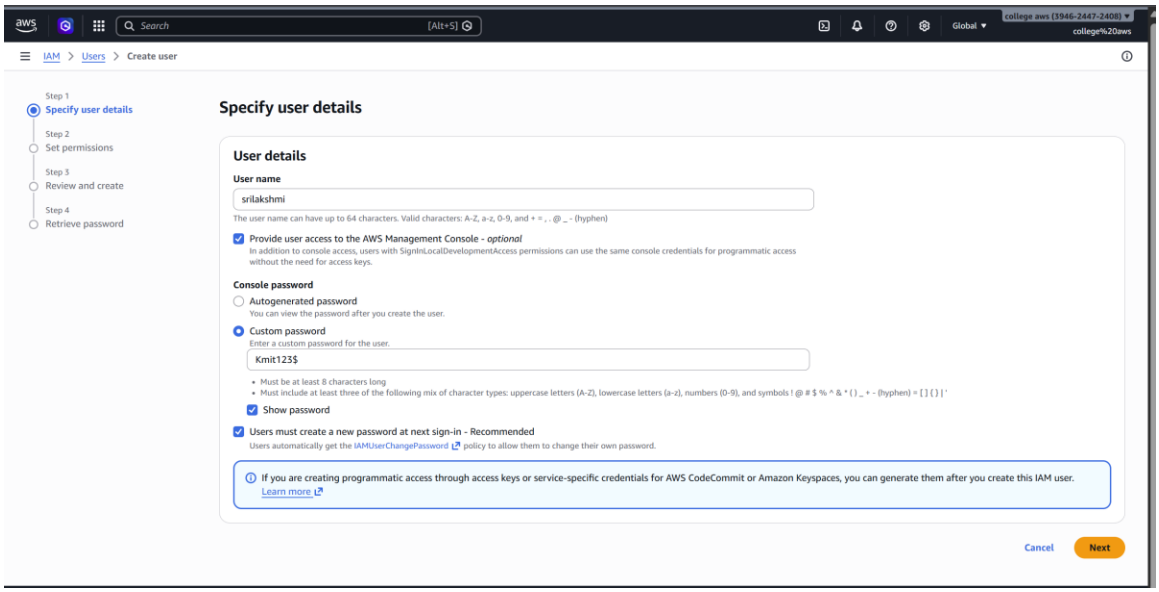
Step 1 – Create IAM User

Login to AWS Console

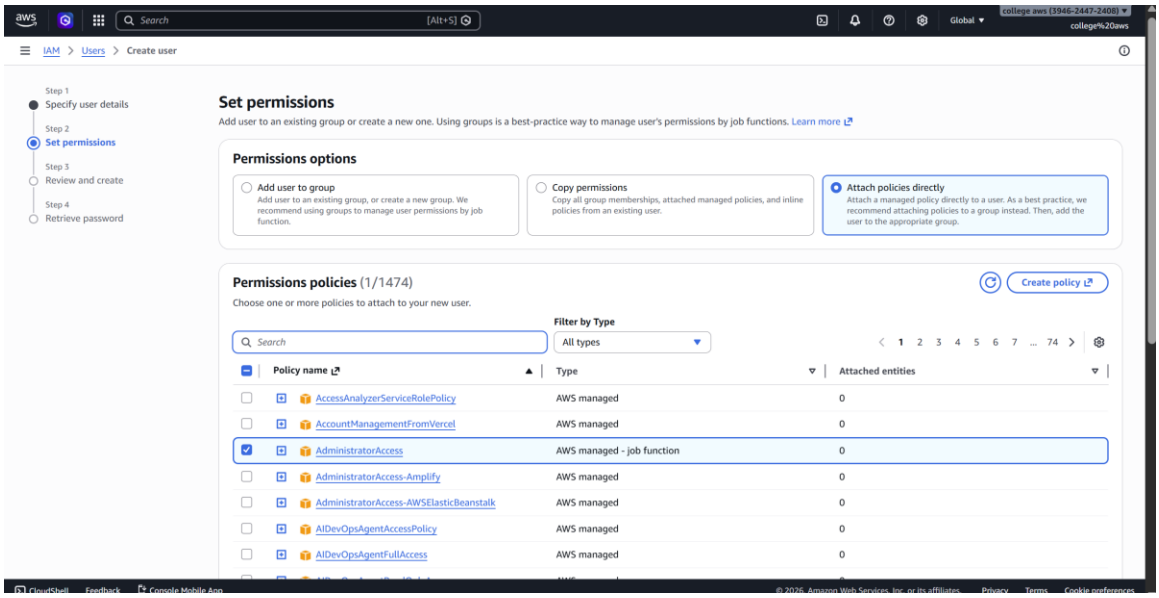
Go to IAM → Users → Create User



Username: lab-user



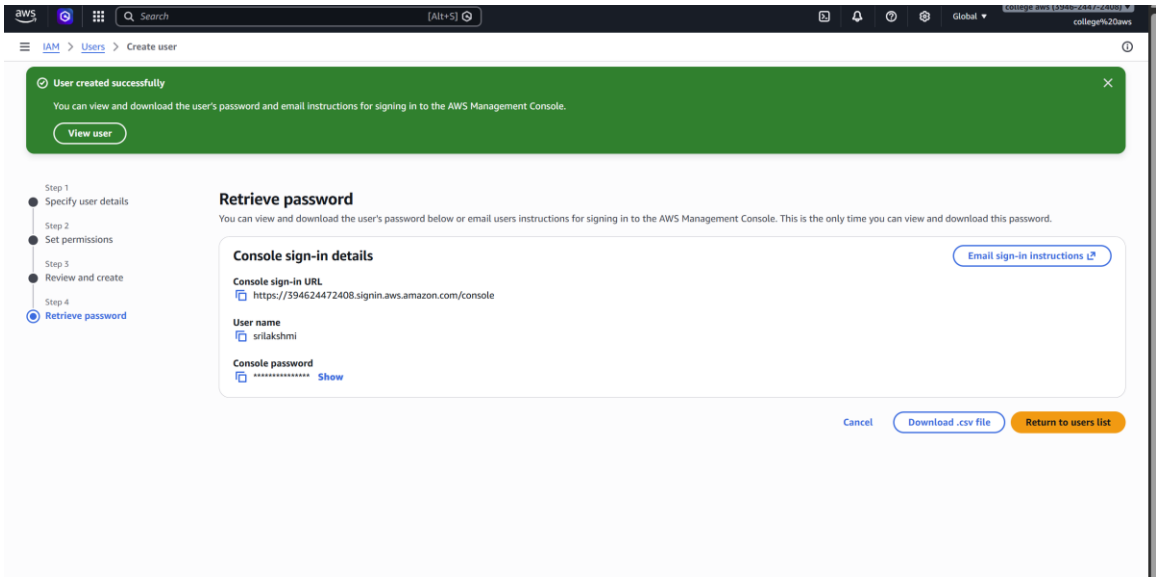
Attach Policy: Administrator Access



Enable Console Login

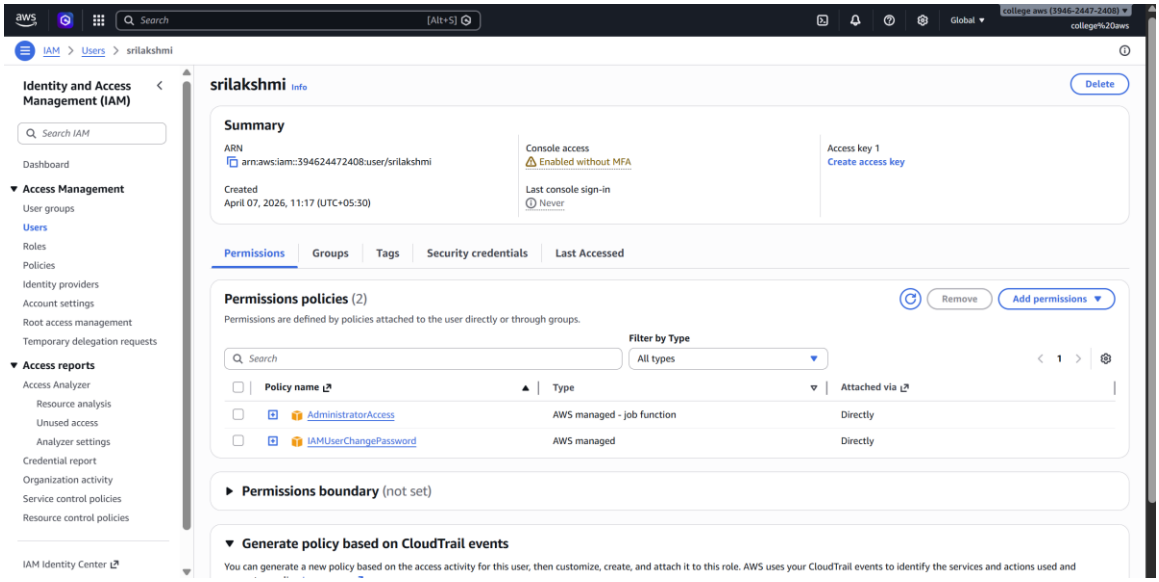
Login using IAM user

IAM User Created

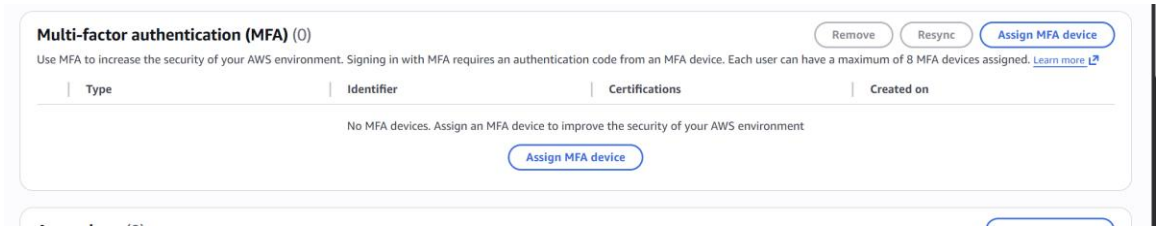


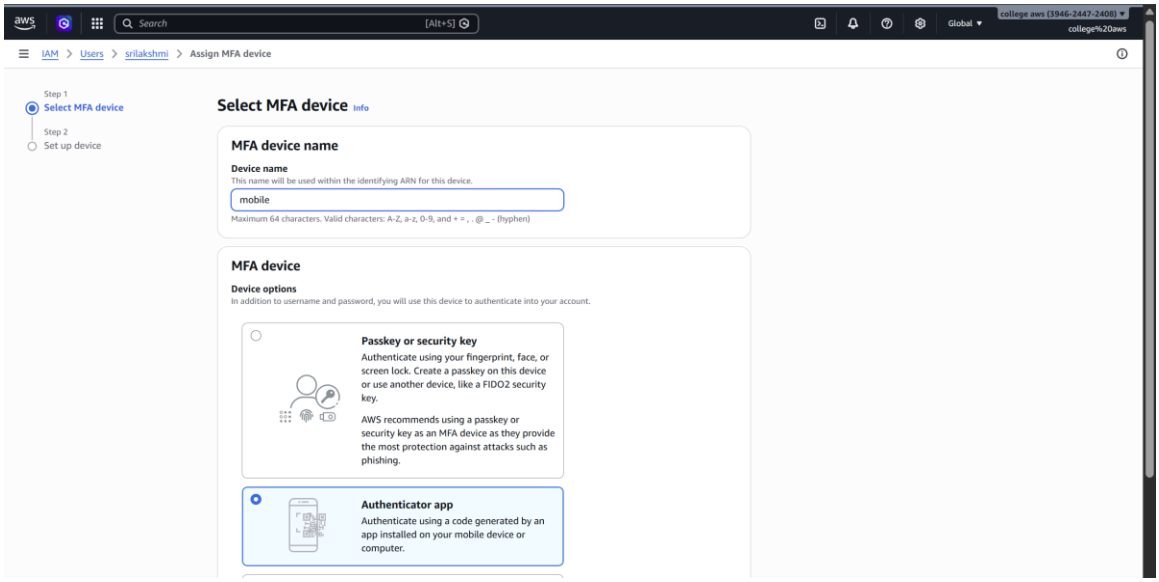
Step 2 – Enable MFA

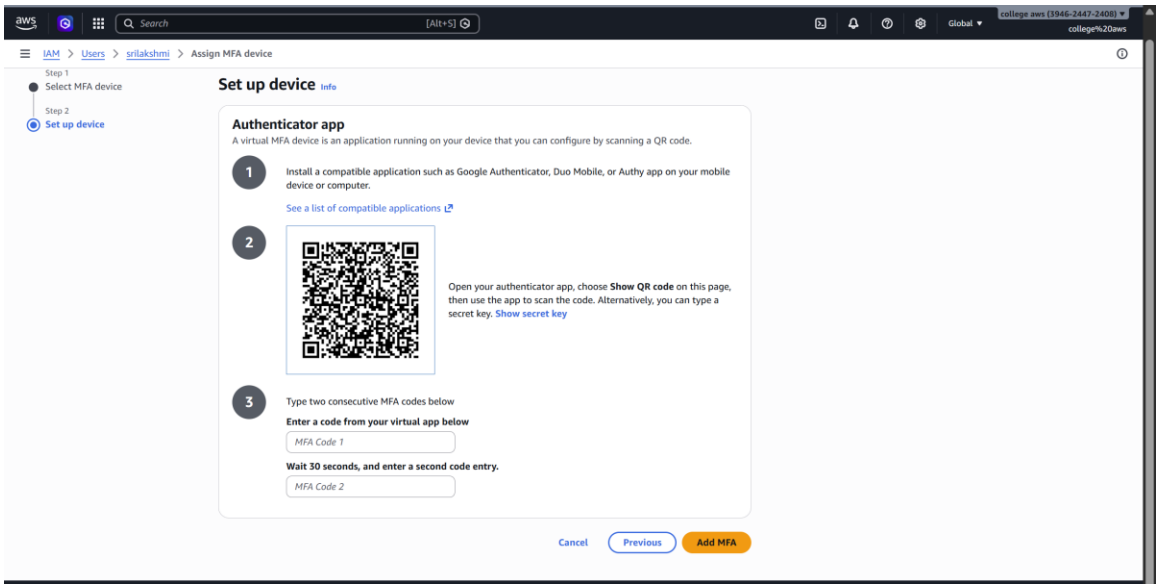
Go to IAM User → Security Credentials



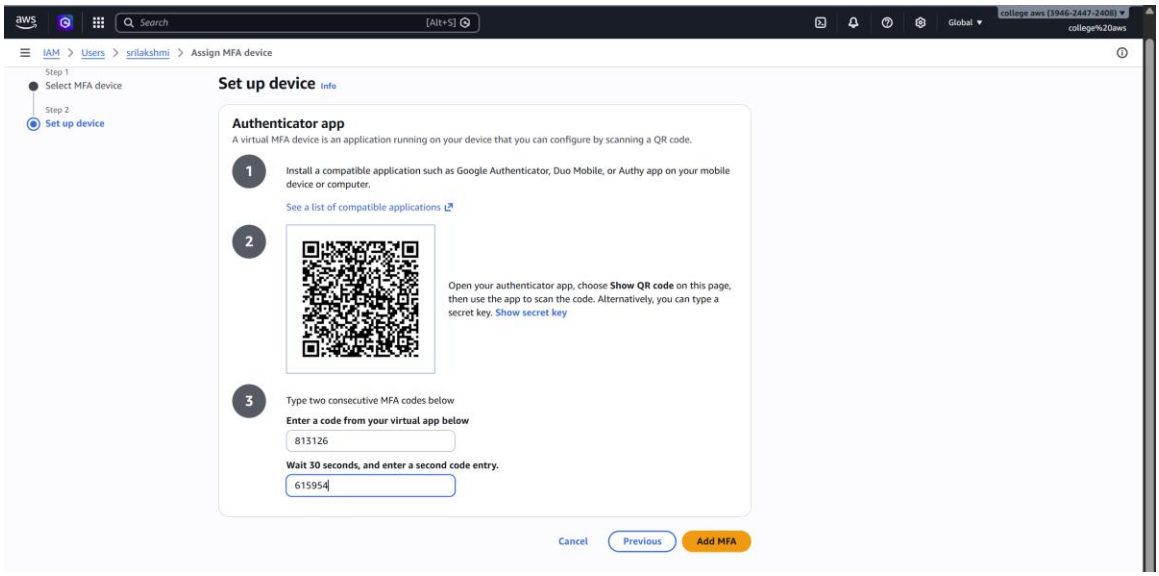
Click Assign MFA





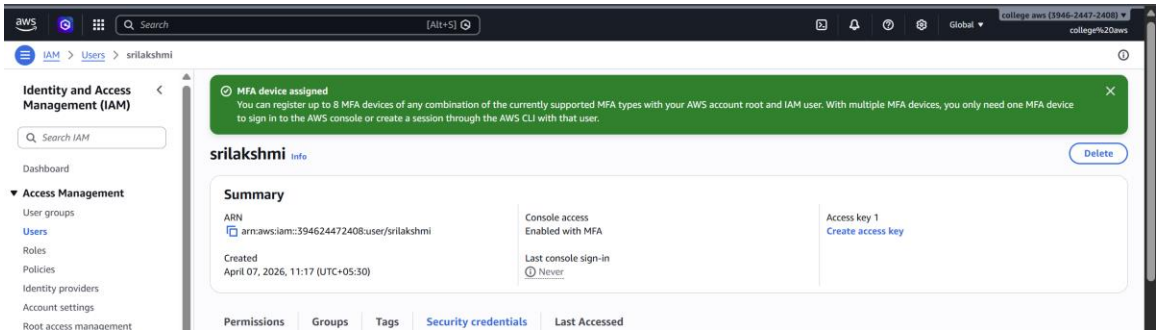


Scan QR code using Google Authenticator



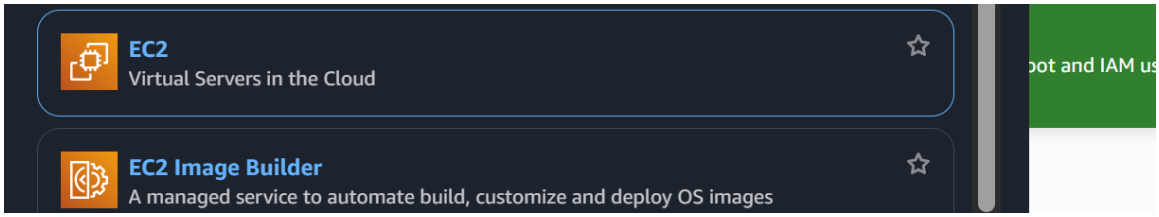
Enter two MFA codes

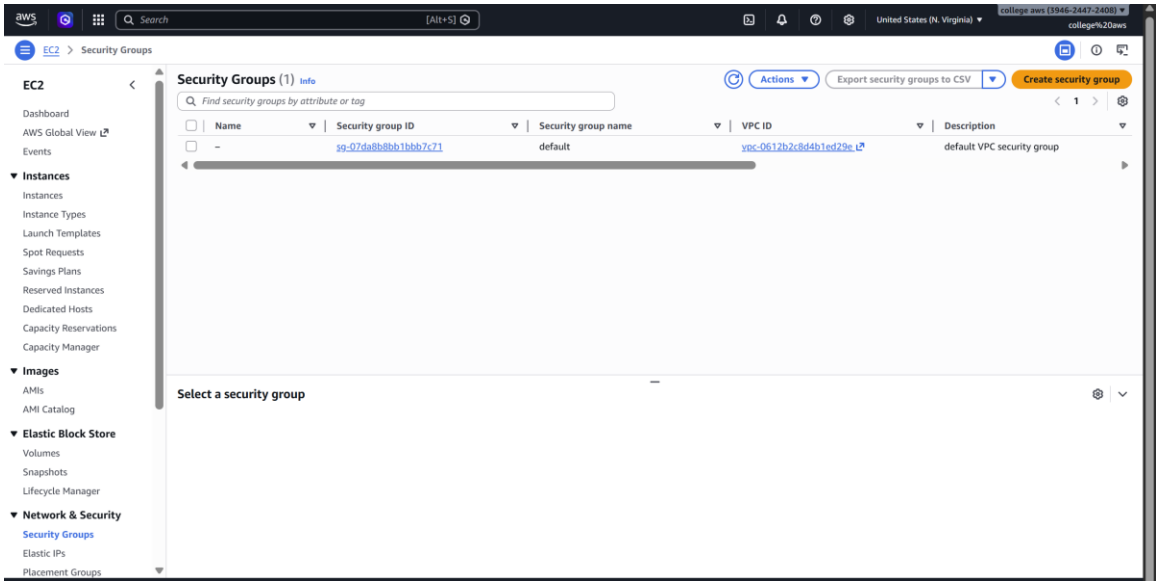
MFA Enabled



Step 3 – Create Security Group

Go to EC2 → Security Groups → Create

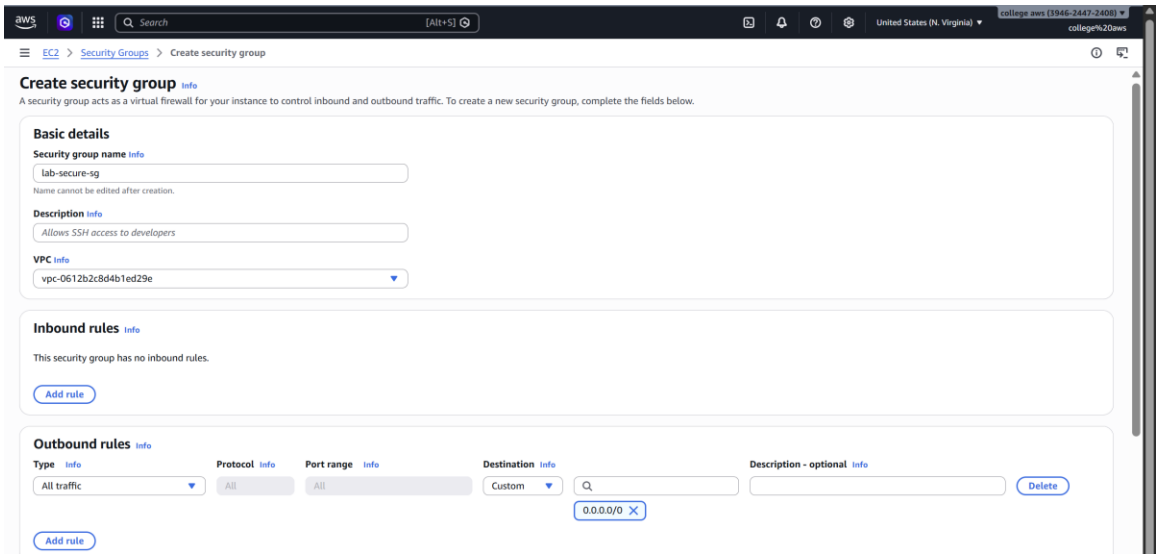




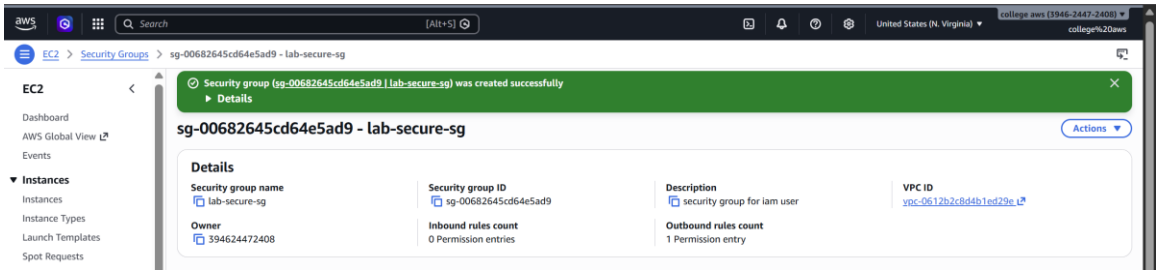
Security Group Name: lab-secure-sg

Inbound Rule: SSH, Port 22, My IP

Outbound: Allow All

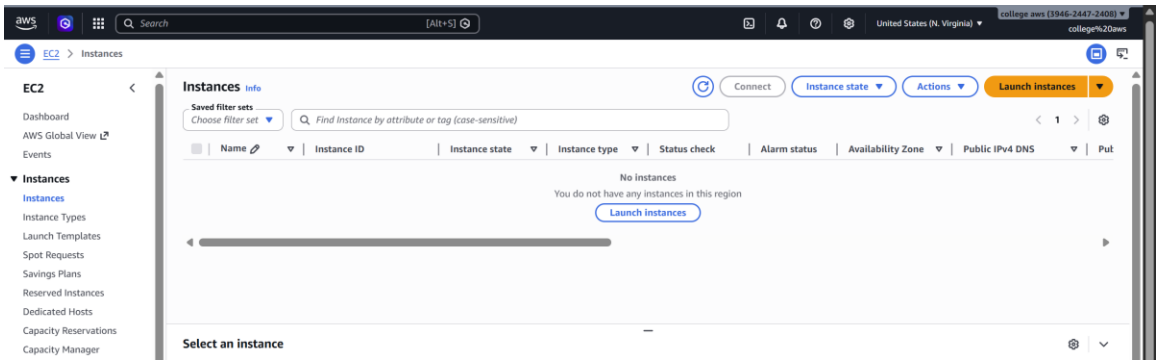


Security Group Created

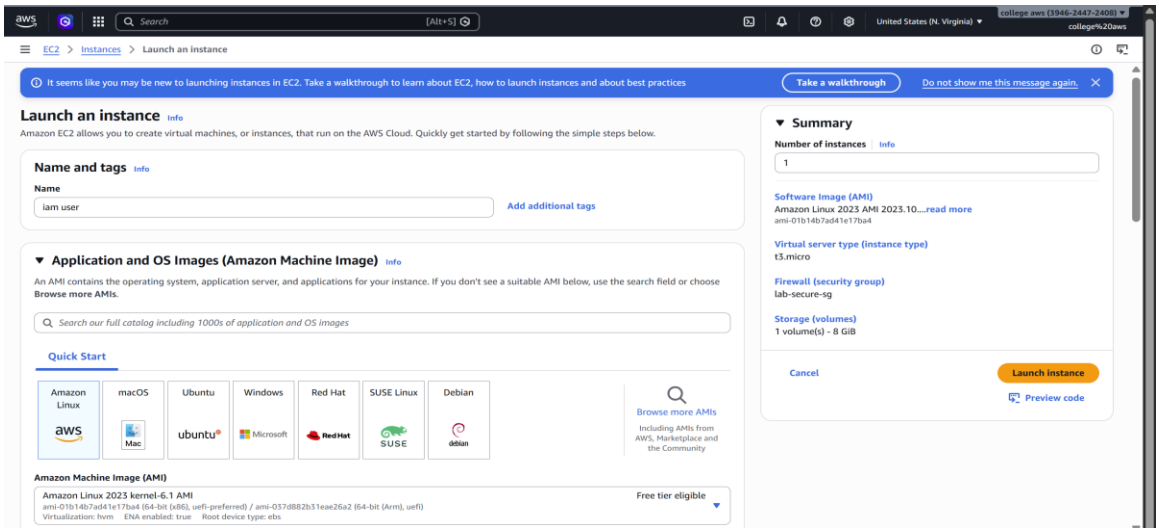


Step 4 – Launch EC2

Go to EC2 → Launch Instance



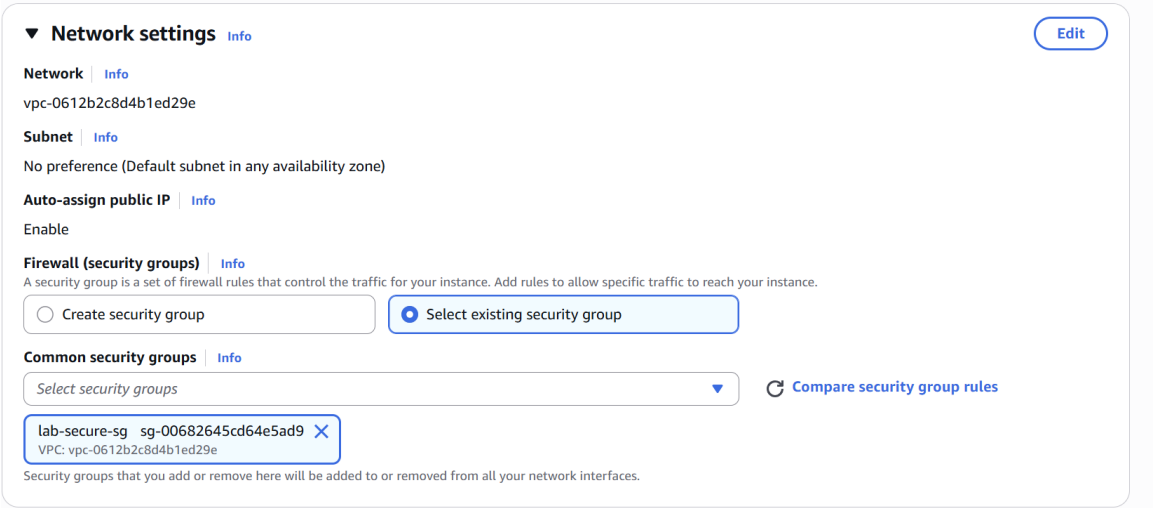
Select Amazon Linux AMI



Instance Type: t2.micro

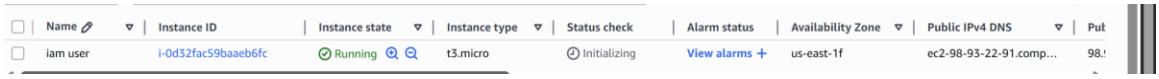
Create Key Pair

Select Security Group



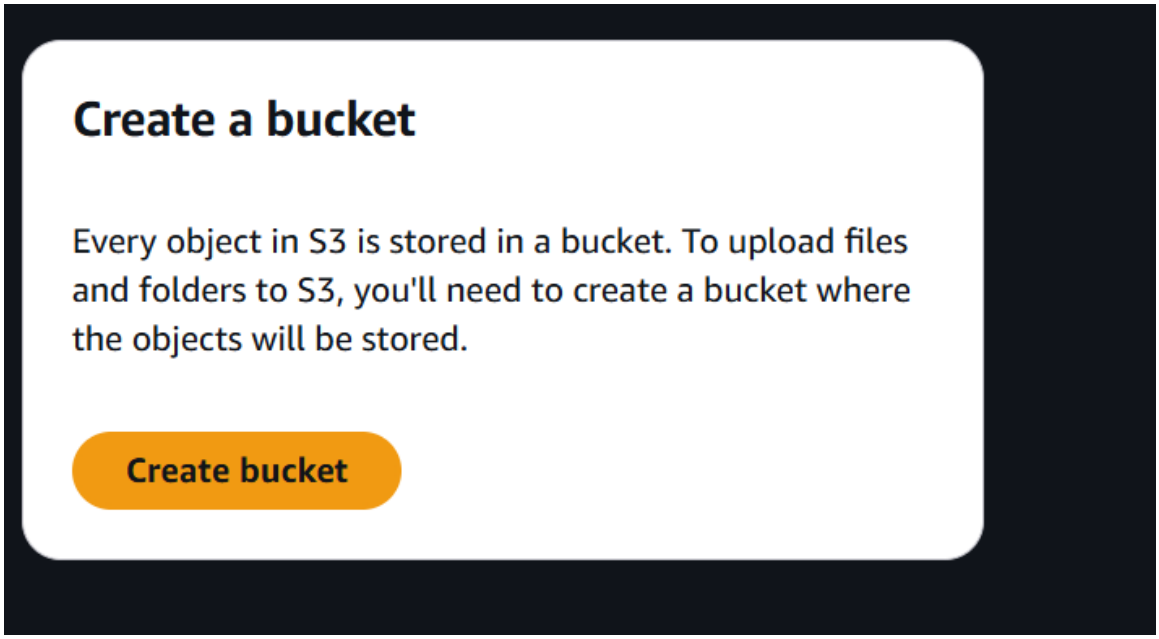
Launch Instance

EC2 Running

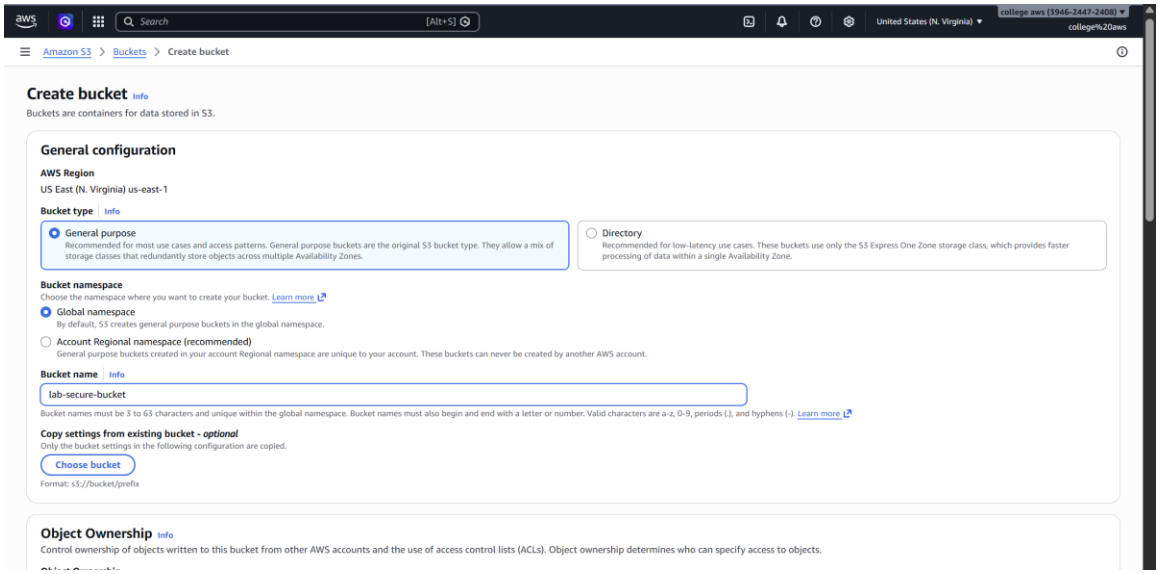


Step 5 – Create S3 Bucket

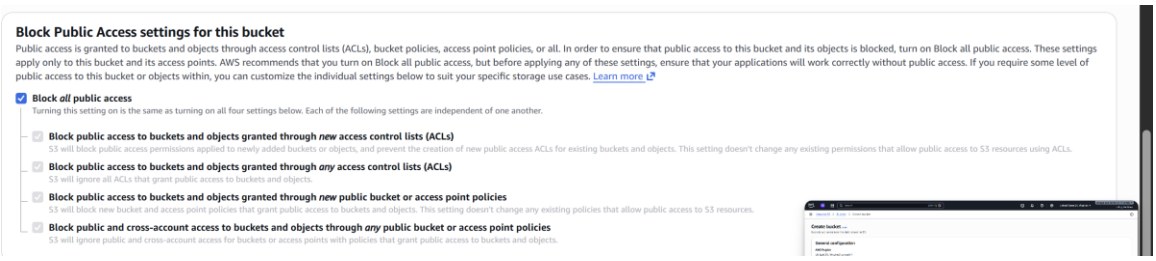
Go to S3 → Create Bucket



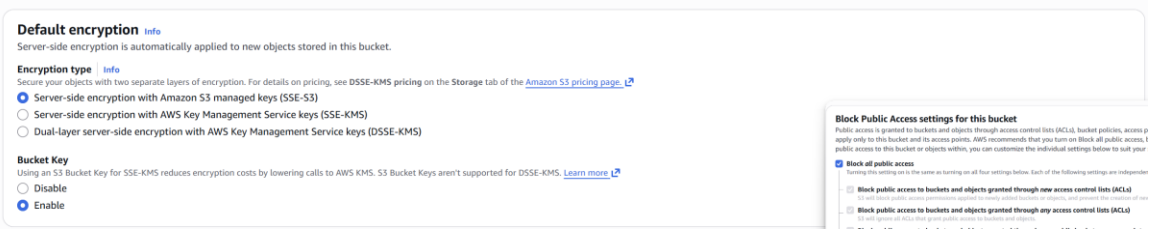
Bucket Name: lab-secure-bucket



Block Public Access: ON



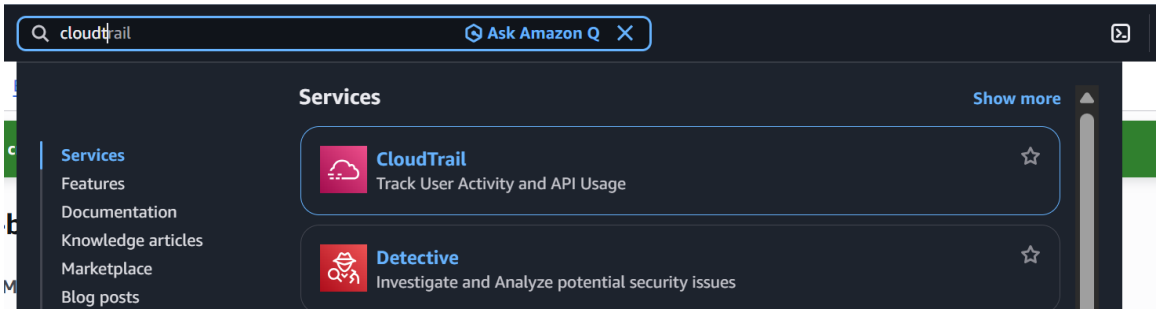
Enable Encryption

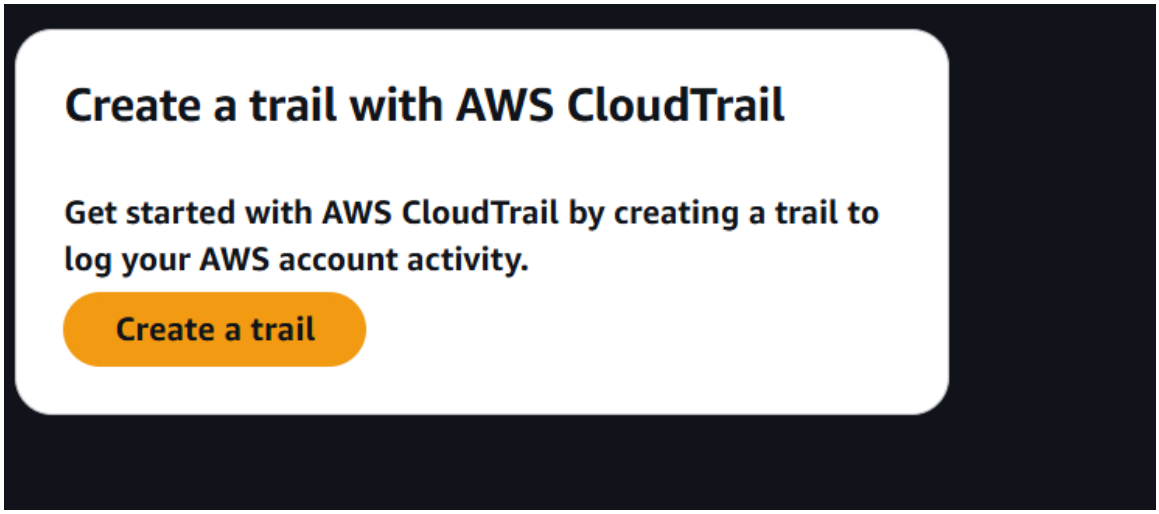


S3 Bucket Created

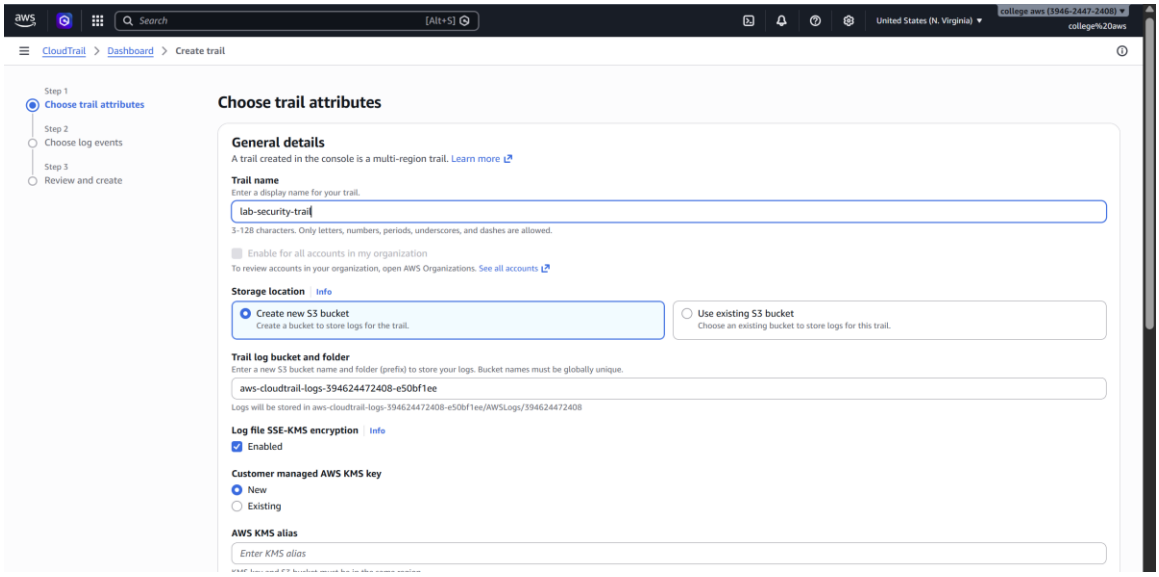
Step 6 – Enable CloudTrail

Go to CloudTrail → Create Trail





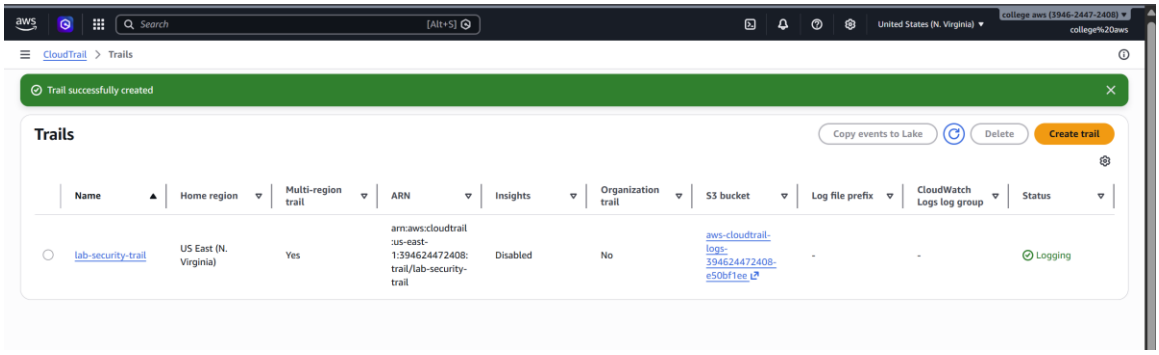
Trail Name: lab-security-trail



Storage: New S3 Bucket

Enable Log File Validation

Logging ON and CloudTrail Enabled



Result

IAM User created

MFA enabled

Security Group configured

EC2 instance launched

S3 bucket secured

CloudTrail logging enabled

Secure cloud setup created successfully